PROBLEM RECAP

Predict booking cancellations

based on customer characteristics and booking details

to help hotels

make smarter operational and pricing decisions

improve

profitability and guest experience

DATASET OVERVIEW

kaggle

Data Source

Hotel Booking Data
Two hotels in Portugal (City & Resort)
from July 2015 to August 2017

Data Dimensions

119,390 rows (bookings)
32 columns (features)

Target Booking Status Cancelled (1) | Not Cancelled (0)

Booking Attributes

lead time
arrival date (y/m/w/d)
days stay (wday/wend)
average daily rate
length of stay
reservation status
average daily rate
days in waitlist
deposit type
reservation status
reservation status date
distribution channel
agent
company

Customer Attributes

# of adults
# of children
# of babies
country of origin
customer type
market segment

Hotel/Room Details

hotel type
meal preference
reserved vs assigned
# of parking
# of special request

Booking Behavior

previous cancellation
repeated guest
booking changes

CHALLENGES & CONCERNS

DUPLICATES
- Managed to keep real values

DATE NOT THE LATEST
- Findings may remain applicable
- Still the most recently available comprehensive full data

OUTLIER & EXTREME
- Transformed and Removed
- Binning

MULTICOLLINEARITY
- Drop features that will cause it

NO PRIVACY CONCERNS
- Customer identification data not included

DATA CLEANING & PREPARATION

ROW/ INSTANCE

raw data

119,390
rows

raw data 119,390
duplicates 31,994
87,396

actual true records

87,396
more accurate & representative

managed missing

-0
mode imputing
drop identifiers

data impute
#children (4)
country (488)

drop feature
agent (12,193)
company (82,137)

outliers handled

-2,384
deleted
impossible values

final data

85,012
97.27%
of actual true records

DATA CLEANING & PREPARATION

FEATURE/ COLUMN

START
32

derive variables **+6**
ex. county → local/domestic

drop variable **-2**
agent
company

adjust type
dates

recode & bin **+11**
ex. #babies → havebaby

BASE
47
FEATURES
possible
multicollinearity

CANDIDATE
18
FEATURES

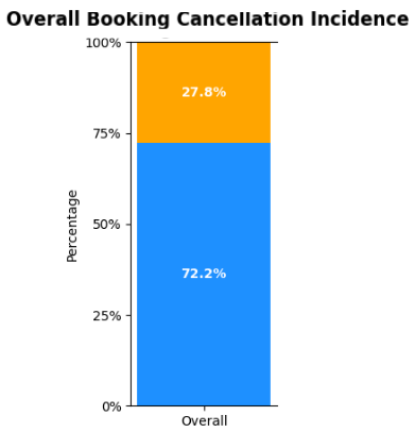
FEATURE DESCRIPTION & DATA TYPE

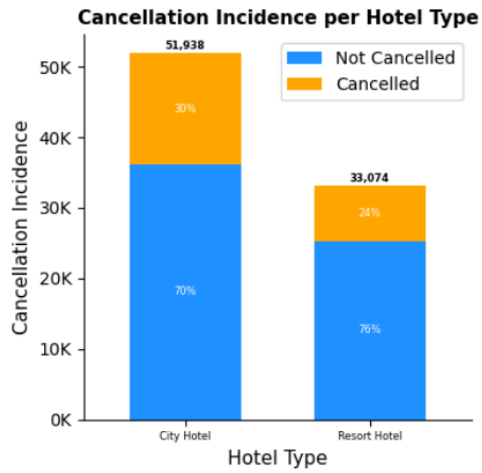
NUMERICAL	Values
Average Daily Rate	USD
INTEGER (BINARY)	
Booking Status	1: cancelled, 0: not cancelled
Weekend	1: yes, 0: no
Repeated Guest	1: yes, 0: no
Waitlisted	1: yes, 0: no
Has Kids	1: yes, 0: no
Guest Country of Origin	1: yes, 0: no
Parking Needs	1: yes, 0: no
Has a Request	1: yes, 0: no
CATEGORICALS	
Arrival Month	Jan, Feb, Mar... Oct, Nov, Dec
Market Segment	OnlineTA, Offline, Direct, Group, Corp, Comp
Hotel	city, resort
Reserved Room Type	A,B,C,D,E,F
Assigned Room Type	A,B,C,D,E,F
Meal	B&B, HalfBoard, FullBoard, None
CATEGORICAL BINS	
Lead Time	1-7, 8-30
Total Stay	0, 1, 2, 3-4, 5-7, 8+
If Cancelled Previously	0, 1, 2+
Booking Change	none, few, several
Number of Guests	1, 2, 3-4, 5+

BOOKING ATTRIBUTES

~1 out of 4 bookings are cancelled

27.8% bookings are cancelled



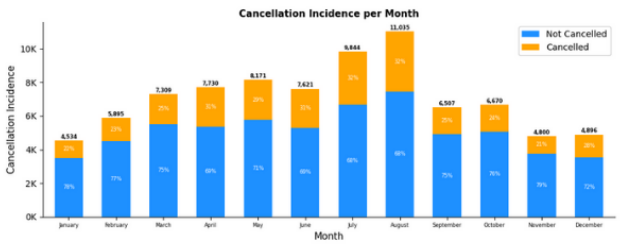


Higher cancellations on city Hotels

30% vs 24%

peak summer bookings peak cancellations

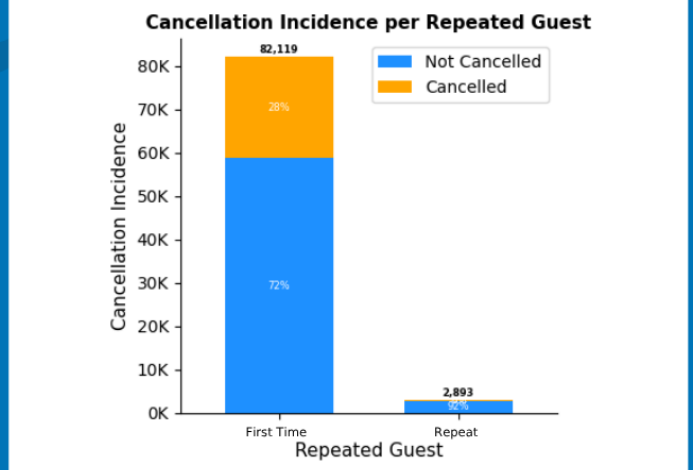
High-demand summer (June-August) months led to more booking, but are also volatile to cancellations



CUSTOMER ATTRIBUTES

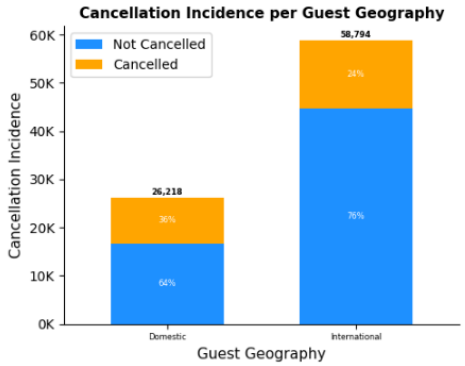
Higher Cancellations among Domestic Guests

36% Among Domestics vs
24% Among Internationsl



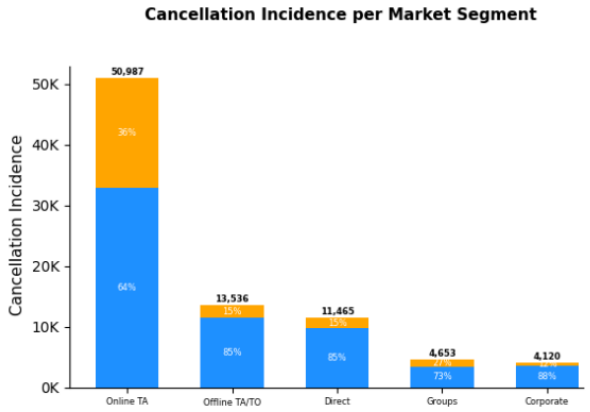
Online Travel Agencies Cancels Most

8% cancellations for repeat guests,
28% rate for new customers



Repeat Customers Tend not to Cancel

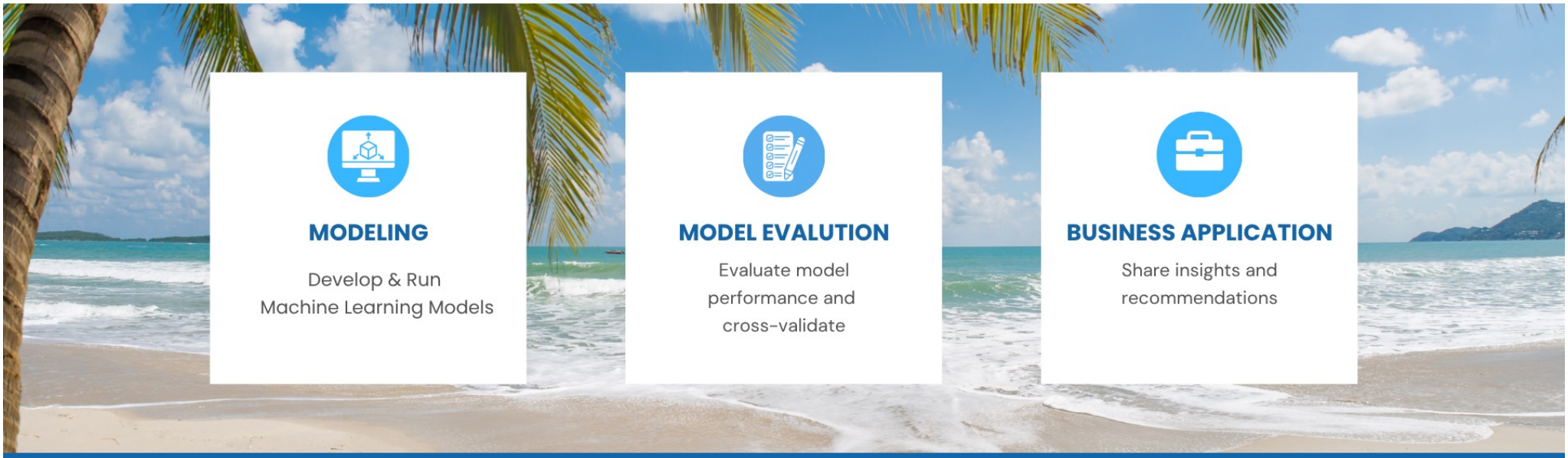
8% cancellations for repeat guests,
28% rate for new customers



VARIABLES INFLUENCING CANCELLATIONS

- Previous Cancellations
- Arrival Month
- Domestic/International
- Hotel Type
- Deposit type
- Booking Segment
- Lead Time
- Number of Guests

NEXT STEPS



MODELING

Develop & Run
Machine Learning Models

MODEL EVALUTION

Evaluate model
performance and
cross-validate

BUSINESS APPLICATION

Share insights and
recommendations

insightout

Decoding Data from Insights to Impact

APPENDIX

DATA CLEANING & PREPARATION

ROW/ INSTANCE

raw data

119,390
rows

raw data 119,390
duplicates 31,994
87,396

actual true records

87,396
more accurate & representative

missing values

-0

data impute
#children (4)
country (488)

drop feature
agent (12,193)
company (82,137)

outlier handling

-2,384

Ave Daily Rate	2178
No of Adults	206
TOTAL	**2384**

final data

85,012
97.27%
of actual true records

DATA CLEANING & PREPARATION

FEATURE/ COLUMN

START
32

feature derive → **+5**

country → **continent**
country → **domestic/int'l**

weekend+weekday=**total days**
adult+children+babies=**total # guest**
children+babies=**total # kids**

log transformation
+1

adr → **log_adr**

feature drop → **-2**

missing values
agent (12,193)
company (82,137)

type adjust → **binary coding** → **+6**

date → isdate
month → categorical

#ofparking → **parkingneed**
#ofparking → **hasrequest**
#baby → **hasbaby**
#kids → **haskids**
stays_in_weekend→ **haveweekend**
#days_in_waiting_list → **is_waitlisted**

numeric binning
+5

#total_days → **total_days_bin**
#lead_time → **lead_time_bin**
#guest_total → **guest_bin**
#ofbookingchanges→**booking_changes_bin**
#ofpreviouscancellations→**prev_cancel_bin**

FINAL BASE
47
FEATURES

FINAL
18
FEATURES

dropping
possible
multicollinearity
candidates